

EXPERIMENT 28:

```
trish@Trisha-PC: ~  
GNU nano 8.7.1 Exp28.cpp  
#include <systemc.h>  
  
SC_MODULE(counting_semaphore) {  
  
    int sem;  
  
    void task() {  
        while(true) {  
            wait(1, SC_SEC);  
  
            if(sem > 0) {  
                sem--;  
                cout << "Task Accessing Resource at " << sc_time_stamp() << " Remaining: " << sem << endl;  
  
                wait(2, SC_SEC);  
                sem++;  
            }  
        }  
    }  
  
    SC_CTOR(counting_semaphore) {  
        sem = 2;  
        SC_THREAD(task);  
        SC_THREAD(task);  
    }  
};  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo      M-G Copy
```

```
trish@Trisha-PC:~$ nano Exp28.cpp  
trish@Trisha-PC:~$ g++ -o Exp28 Exp28.cpp -lsystemc  
trish@Trisha-PC:~$ ./Exp28  
  
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46  
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ALL RIGHTS RESERVED  
  
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0  
In file: kernel/sc_object_manager.cpp:153  
  
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1  
In file: kernel/sc_object_manager.cpp:153  
Task Accessing Resource at 1 s Remaining: 1  
Task Accessing Resource at 1 s Remaining: 0  
Task Accessing Resource at 3 s Remaining: 0  
Task Accessing Resource at 4 s Remaining: 0  
Task Accessing Resource at 5 s Remaining: 0  
Task Accessing Resource at 6 s Remaining: 0  
Task Accessing Resource at 7 s Remaining: 0  
Task Accessing Resource at 8 s Remaining: 0  
Task Accessing Resource at 9 s Remaining: 0  
trish@Trisha-PC:~$
```